Enhanced Rock Weathering for Sustainable Carbon Capture: Advances, Challenges, and Future Prospects in Terrestrial Ecosystems

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Abstract

Enhanced Rock Weathering (ERW) is a carbon dioxide removal strategy with the dual promise of sequestering atmospheric carbon and improving agricultural soils. This review critically assesses ERW's real-world effectiveness, emphasizing the need for evidentiary standards based on verifiable flux closure and permanence of carbon storage. We synthesize the complex interplay of geochemical kinetics, hydrology, mineralogy, particle size, and biological activity, concluding that the wide variance in reported field-scale removal rates (from approximately 0 to 9 tonnes of CO_2 per hectare per year) stems from site-specific environmental conditions rather than intrinsic mineral properties alone. Consequently, common proxies like soil pH changes or crop yield gains are often poor indicators of net removal. Lifecycle analysis reveals that energy for grinding and logistics are dominant costs and emissions, suggesting an optimal particle fineness for net efficiency. A tiered modeling appraisal highlights the risk of overestimation in current models, with plausible 2050 deployment potential estimated at 0.2–4 gigatonnes of CO_2 per year. We also evaluate significant environmental and social risks, including trace-metal accumulation, dust exposure, and equity concerns. The review concludes by proposing a framework for credible scaling that integrates robust monitoring, multi-scale modeling, and safety-conscious policy, positioning ERW as a meaningful contributor within a diversified carbon removal portfolio when rigorously implemented.

1. Introduction

The Earth's carbon cycle on a geological time scale is intricately linked to terrestrial weathering and the formation of carbonate minerals in ocean, a process that acts as a natural regulator of atmospheric carbon dioxide (CO_2) concentrations (Walker et al. 1981; Berner 1994). With the specter of climate change looming large, the scientific community has intensified its pursuit of carbon capture and storage solutions, particularly through the lens of enhanced rock weathering (ERW)—a geoengineering strategy to expedite the naturally occurring mineral weathering to sequester atmospheric CO_2 (Berner, 1992; Ludwig et al., 1999; Seifritz, 1990). In the scope of this article, we define ERW as a spectrum of methodologies designed to accelerate the carbonation and enduring sequestration of minerals within farmland or natural terrestrial ecosystems, proceeding at an expedited rate relative to the baseline weathering of untouched source minerals. While there is burgeoning interest in the potential for

carbon sequestration within synthetic materials such as concrete, these ventures, though valuable, fall outside the purview of ERW as discussed in this review.

Initial investigations into ERW were largely confined to understanding the basic geochemical reactions involved in the natural weathering of rocks and how these processes could be accelerated to capture more CO_2 from the atmosphere (Caldeira 1995; Kojima et al. 1997; Gaillardet et al. 1999). Insights were gained into the geochemical mechanisms, those influence rock weathering, establishing a theoretical basis for future research (Matsuoka 1995). Research on ERW progressed by exploring the potential of various rock types for carbon sequestration. In the early 2000s, studies began to emphasize the significant role of naturally occurring shales and basalts on CO_2 uptake through geochemical weathering, providing a new avenue for understanding the role of different rock types in carbon sequestration (Amiotte Suchet et al. 2003). By 2014, the focus shifted towards optimizing the CO_2 efficiency of enhanced weathering, with researchers highlighting the importance of selecting appropriate source rocks and material comminution techniques (Moosdorf et al. 2014). This period also saw a growing interest in understanding how the interactions between water, CO_2 and minerals in aquatic ecosystems contribute to the carbon sinks produced by rock weathering (Liu and Dreybrodt 2015). Towards the end of the last decade, the potential for large-scale CO_2 removal through ERW started to gain prominence. Notably, Beerling et al. (2018) provided a detailed assessment of the techno-economic potential of implementing ERW on croplands in the UK, including the associated costs and engineering challenges. Beerling et al. (2020) have further argued that ERW stands out in the climate mitigation landscape by presenting a detailed sub-national level analysis of its cost-effectiveness and atmospheric carbon dioxide removal (CDR) efficacy.

In recent years, research on ERW has markedly evolved, delving deeper not only into the geochemical efficiency of carbon capture (Deng et al. 2023; Knapp et al. 2023; Ryan et al. 2024; Epihov et al. 2024) but also embracing broader environmental sustainability (Feng and Hicks 2023; Dupla et al. 2023; Bi et al. 2024) and social equity dimensions (Cox et al. 2022; Sovacool et al. 2023; Low et al. 2024). The emphasis on environmental sustainability underscores the critical role ERW plays in promoting a balanced ecosystem and mitigating the adverse effects of climate change (Taylor et al. 2021). A critical aspect of ERW is that it has potential to yield societally positive benefits that are not directly related to mitigating climate change. For example, in appropriate contexts, ERW can revitalize soil quality (te Pas et al. 2023; Skov et al. 2024), enhance soil fertility (Bi et al. 2024), reduce erosion (Edwards et al. 2017), and fostering biodiversity within the soil ecosystem (Hicks et al. 2022). As the potential environmental benefits of ERW became more evident, researchers also started to delve into ERW's societal implications. Studies have highlighted ERW's versatility, showcasing its potential applications in diverse settings from urban farming (Haque et al. 2021) to traditional agricultural landscapes (Swoboda et al. 2022; Buckingham and Henderson 2024) In coastal regions, there have been focused studies on the synergy between ERW and marine ecosystems, investigating how ERW can provide co-benefits such as enhancing carbon sequestration in ocean waters and improving marine biodiversity (Kanzaki et al. 2023; Foteinis et al. 2023). ERW research has also broadened to include infrastructure development, revealing promising avenues for integrating ERW into sustainable urban planning and construction (Jorat et al. 2022).

While ERW holds promise as a CDR method, significant knowledge gaps remain. These gaps encompass various aspects, including the intricate factors influencing long-term effectiveness of ERW, environmental consequences, scalability prospects, and economic feasibility. Additionally, the complex interplay between ERW and local ecosystems, as well as its potential side effects on soil health and water resources, requires further investigation. This review thus embarks on a comprehensive examination of ERW, delving into the foundational geochemical dynamics such as carbonation reactions, the modulation of alkalinity, and the nuanced implications of particle size and mineral armoring on the efficacy of the process. We critically explore the geochemical and environmental underpinnings of ERW, assessing how different rock types and abiotic factors influence its viability. The examination extends to uncover the biological synergies of ERW, examining the integral roles of microbial activity, plant roots, and soil fauna in facilitating carbon sequestration and their impact on soil nutrient cycles and terrestrial ecosystem carbon cycles. Further, this review contrasts observational data with economic and policy considerations to evaluate ERW's practicality and scalability. We also consider modeling perspectives to understand ERW's complexities and predict its outcomes. This review further compares ERW with other CDR techniques to position it within the climate mitigation landscape. We identify and discuss critical knowledge gaps, from geochemistry and experimental data to environmental safety and socio-economic integration, aiming to foster a discerning advancement of ERW that is environmentally, economically, and socially equitable.

2. Foundational Geochemical Dynamics of ERW

The geochemical dynamics of ERW matter only insofar as they deliver persistent carbon removal with verifiable flux closure. Figure 1 correctly situates reaction pathways alongside biophysical interactions and grain size effects.

Yet its implications become decision relevant only when three links are explicit. First, alkalinity produced by silicate dissolution must be traced to export and to durable oceanic or pedogenic sinks rather than inferred from stoichiometry alone. Second, reactive surface area evolves in time because of transport, passivation, and biological modification, so any single "rate constant" is context dependent. Third, the coupling between reaction and transport controls whether laboratory kinetics translate into field-scale alkalinity flux. The sections below interrogate these links using the evidence summarized in Table 1 and Table 2, and they identify testable claims that can be falsified by field measurements.

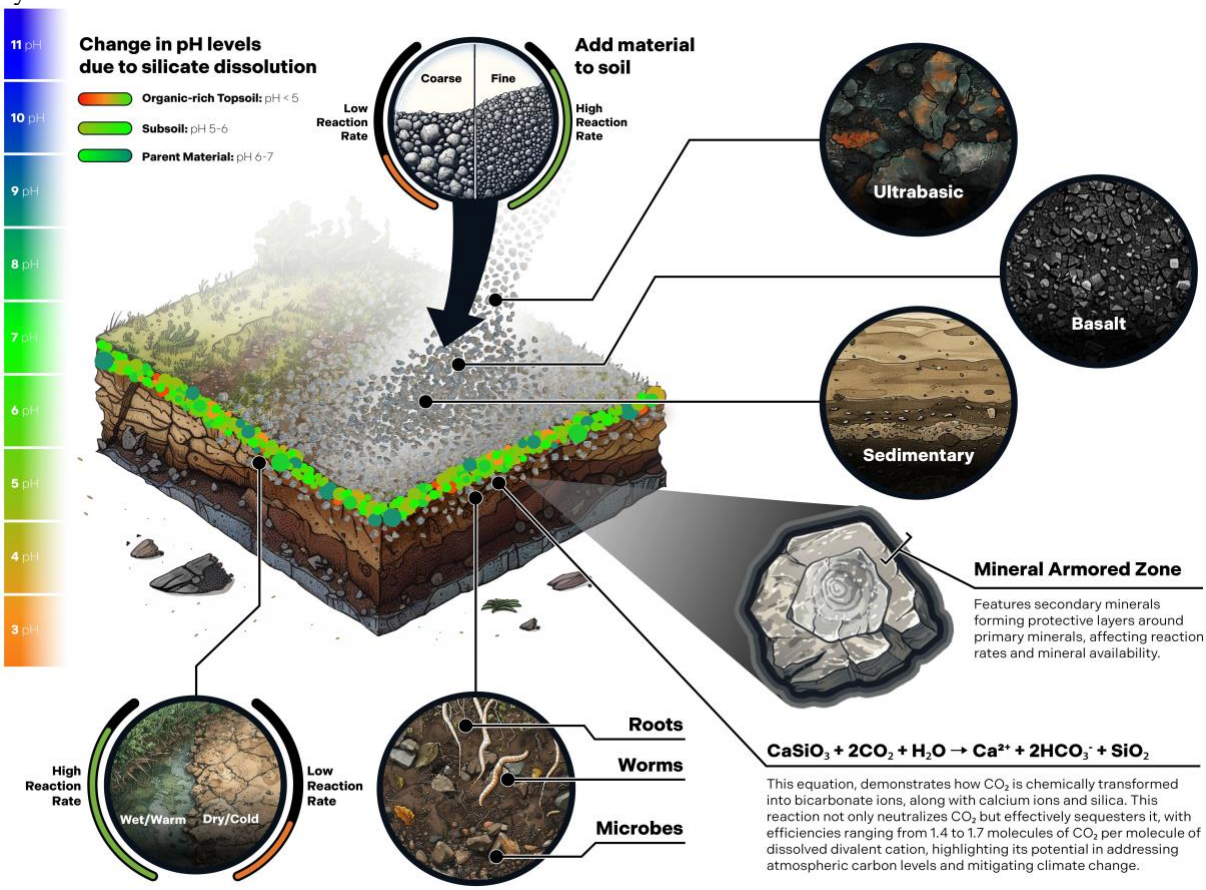


Figure 1. Geochemical Dynamics and Biophysical Interactions in Enhanced Rock Weathering. This infographic depicts the Enhanced Rock Weathering (ERW) process as a strategy for mitigating climate change. It showcases how various rock types (e.g., ultrabasic, basalt, sedimentary) are processed into coarse and fine particles and integrated into soil to speed up natural weathering reactions. These reactions facilitate the breakdown of silicate minerals by absorbing CO_2 from the atmosphere, converting it into bicarbonate ions, which are subsequently washed into the ocean, permanently removing CO_2 from the air. The diagram illustrates pH changes across different soil layers (e.g., organic-rich topsoil, subsoil, and parent material), highlighting the potential impact of the weathering process on soil acidity. It also underscores the role of soil microbes, plant roots, and soil fauna in enhancing mineral breakdown, varying by environmental conditions (wet/warm vs. dry/cold). Furthermore, it depicts how smaller particle sizes increase reaction rates due to greater surface areas. The infographic also presents a mineral armored zone that includes secondary minerals forming protective layers around primary minerals, influencing their availability and the reaction rates. The infographic concludes with the geochemical equation that illustrates the transformation of silicate minerals into bicarbonates, emphasizing the sequestration of atmospheric CO_2 and its implications for climate mitigation. The soil profile depicts a general pH gradient increasing with depth, with color variations indicating natural heterogeneity within layers.

2.1 Reaction Mechanisms

In the context of ERW for CO_2 sequestration, the primary mechanism considered is silicate mineral weathering (Hartmann et al. 2013). While carbonate weathering also sequesters CO_2 over short timescales (e.g., less than 10^5

years, Zhang et al. 2022), its long-term net effect on atmospheric CO_2 is close to zero. Therefore, we limit the discussion to silicate weathering in this review.

Silicate minerals are naturally occurring compounds composed of silicon and oxygen, often with additional cations. These minerals undergo reactions with CO_2 and water that result in the release of cations (e.g., Ca^{2+}, Mg^{2+}):

$$\text{Silicates} + CO_2 + H_2O \rightarrow \text{Cations} + HCO_3^- + SiO_2 \quad (1)$$

On a decadal scale, silicate weathering sequesters CO_2 primarily by increasing the alkalinity of soil and aquatic environments through the formation of bicarbonate (HCO_3^-) and carbonate (CO_3^{2-}) ions (Hartmann et al. 2013). The released cations are partially stored in exchangeable sites in soils, precipitated as pedogenic carbonates, and predominantly released to form carbonate minerals (e.g., calcite, dolomite) that deposit on the ocean floor in various rock forms (Hartmann et al. 2013). This results in the net reaction:

$$\text{Silicates} + CO_2 + H_2O \rightarrow \text{Carbonates} + SiO_2 \quad (2)$$

Dissolution of silicate minerals is a crucial geochemical process in ERW. The rate of silicate dissolution (r, mol s^{-1}) has been extensively studied (see (Brantley 2008). for a review) and is typically formulated as a function of chemical affinity (the distance from equilibrium), specific surface area, solution pH, temperature (T), and mineral types (Palandri and Kharaka, 2004):

$$r = -m \cdot SSA \cdot \sum_I \left\{ k_i \exp\left[-\frac{E_i}{R}\left(\frac{1}{T} - \frac{1}{298.15K}\right)\right] a_i^{n_i}(1-\Omega^{p_i})^{q_i} \right\}$$

$$\Omega = \frac{Q}{k_{eq}}$$

$$Q = \frac{\Pi_m [A_m]^{c_m}}{\Pi_n [A_n]^{c_n}}$$

, where the definitions are:

$i \in \{H^+, H_2O, OH^-\}$ – different weathering agents under acid, neutral, and basic conditions

m – the mass of the mineral, g

SSA – specific surface area, $m^2\ g^{-1}$

k_i – dissolution rate constants at 298.15K, mol $m^{-2}\ s^{-1}$

E_i – Arrhenius activation energy of the reaction, J mol^{-1}

T – temperature, K

R – natural gas constant, 8.31 J K^{-1} mol^{-1}

a_i – activity of the aqueous species i, $a_{H_2O} = 1$ and $a_{H^+}a_{OH^-} = 10^{-14}$

n_i – reaction order with respect to species i, $n_{H_2O} = 1$

Ω – the mineral saturation index

p_i, q_i – empirically dimensionless constants, generally set to 1 due to lack of information

Q – activity product

163 k_{eq} – equilibrium constant for the dissolution reaction

164 $[A_m]$ – concentration of the aqueous species m on the right-hand side of the dissolution reaction equation

165 $[A_n]$ - concentration of the aqueous species n on the left-hand side of the dissolution reaction equation

166 c_m, c_n – stoichiometric coefficients of the dissolution reaction equation

167 The widely used dissolution rate formalism captures dependence on chemical affinity, pH, temperature, and surface
168 area (Brantley 2008). However, the values reported in **Table 1** arise from different reactors, background electrolytes,
169 distances from equilibrium, and area metrics, which complicates direct comparison. As a result, the qualitative trend
170 that less polymerized frameworks dissolve faster than tectosilicates remain robust, while the quantitative spread in
171 rates and apparent activation energies is not directly portable across studies without harmonized conditions
172 (Brantley 2008). In the field, additional controls intervene, including defect-controlled hotspots, evolving silica
173 activities that drive surface passivation, and temperature effects that can increase intrinsic rates while simultaneously
174 pushing solutions toward equilibrium.
175 Material choice therefore embeds a risk–benefit calculus. Ultramafic sources rich in olivine offer higher intrinsic
176 reactivity but also carry Ni, Co, and Cr that can mobilize under weathering and accumulate in plants and soils,
177 which raises monitoring and permitting hurdles (Alexander, 2007; Ten Berge et al., 2012; Amann et al., 2020).
178 Basalts are more widely available and agronomically friendlier, often delivering phosphorus along with alkalinity,
179 although unit-mass CO_2 removal is typically lower than for ultramafic (Middlemost, 1975; Strefler et al., 2018;
180 Beerling et al., 2024; Goll et al., 2021; Lewis et al., 2021). Consequently, claims of superior performance must be
181 anchored in flux closure and co-reported risk indicators rather than in isolated kinetic constants.
182 A practical inference follows. When reporting "removal," studies should provide concurrent measurements of
183 alkalinity export, major cation stoichiometry consistent with silicate dissolution, and fluid residence times. Under
184 these conditions, Table 1 can be used not as a catalogue of numerical rates but as a prior for bounding field-effective
185 rates once hydrologic contact and silica feedbacks are characterized.
186
187 **Table 1.** Dissolution rate constants (Log k, mol m^{-2} s^{-1}) and Arrhenius activation energies (E, kJ mol^{-1}) of selected
188 land ERW minerals under acidic and neutral mechanisms.

Classification of Silicate Minerals		**Formula**	**Acid mechanism**				**Neutral mechanism**			**Study**[a]
			log k	**E**	**n**	**pH**	**log k**	**E**	**pH**	
Orthosilicates	Fayalite (Olivine)	Fe_2SiO_4	-4.80	94.4	1.000	--	--	94.4	--	(Renforth et al. 2015)
	Forsterite (Olivine)	Mg_2SiO_4	-6.85	67.2	0.470	1.03–8.71	-10.64	79.0	9.30–12.06	
Phyllosilicates	Lizardite	$Mg_3Si_2O_5(OH)_4$	-5.70	75.5	0.800	--	-12.40	56.6	--	(Taylor et al. 2016)
Inosilicates	Wollastonite	$CaSiO_3$	-5.37	54.7	0.400	--	-8.88	54.7	--	(Jariwala et al. 2022)
	Tremolite	$Ca_2Mg_5Si_8O_{22}(OH)_2$	-8.40	18.9	0.700	--	-10.60	94.4	--	(Paulo et al. 2021)
	Diopside	$CaMgSi_2O_6$	-6.36	96.1	0.710	--	-11.11	40.6	--	(Lewis et al. 2021)
Tectosilicates	Anorthite	$CaAl_2Si_2O_8$	-3.50	16.6	1.411	--	-9.12	17.8	--	(Baek et al. 2023)

	Albite	$NaAlSi_3O_8$	-10.2	65.0	0.457	1.3–4.0	-12.6	69.8	5.6–8.2	(te Pas et al. 2023)
	K feldspar	$KAlSi_3O_8$	-10.1	51.7	0.500	4–5.91	-12.4	38.0	6–7	(Xiao et al. 2016)

a: The references in the table indicate studies where the minerals were used, while the information on the acid mechanism and neutral mechanism comes from (Palandri and Kharaka 2004).

2.2 Alkalinity Modulation and pH Dynamics

The pH responses in Table 2 span from approximately 3.6 to 8.5 across climates, lithologies, particle sizes, application rates, and exposure durations. This range confirms that ERW can rapidly shift soil acidity, particularly under fine particles, larger inputs, and longer treatments (Gillman et al., 2002; Anda et al., 2009; Anda et al., 2013; Kelland et al., 2020; Amann et al., 2020; Ten Berge et al., 2012; Dietzen et al., 2018; Johnson et al., 2014a; Linke et al., 2024; Ning et al., 2016). Nevertheless, pH is a state variable rather than a flux. A short-term rise can reflect occupation of exchange sites or proton consumption that does not translate into alkalinity export. Therefore, pH trajectories must be interpreted together with bicarbonate concentrations, cation balances, and outflow rates. Without that closure, inferences about carbon removal remain provisional.

The agronomic narrative is equally contingent. Enhanced pH can improve nutrient availability and microbial activity and can raise yields in some contexts, but the magnitude and persistence of these benefits depend on soil buffering capacity, background acid loads, and management practices (S. Liu et al., 2021; Swoboda et al., 2022; Wood et al., 2023; Guo et al., 2023). Consequently, studies that invoke co-benefits as part of the value proposition should co-report time-resolved pH, alkalinity export, and nutrient budgets to distinguish durable removal from transient conditioning.

Two interpretive priorities emerge for Figure 1 and Table 2. First, distinguish clearly between pH adjustment and alkalinity generation, and attach comparable hydrologic and ionic data to each pH observation. Second, emphasize the feedback whereby pH elevation can slow proton-driven dissolution, since this sets a natural trajectory of early rapid response followed by kinetic dampening.

Table 2. Impact of rock types and locations on soil pH adjustments through ERW

Rock Types	Location	Soil type	Temp	Particle Size (μm)	Application rate (g/m^2)	Duration (month)	pH change	Study
Basalt	Iceland	Andosol	7 °C	10 to 62	500–800	7	4.4 to 6	(Linke et al. 2024)
Basalt	Australia	Oxisols	Room temp	<150	2,500	3	4.75 to 5	(Gillman et al. 2002)
Basalt	Malaysia	Oxisols	26–28°C	<250	500–8,000	3–15	3.9 to 5.1	(Anda et al. 2009)
Basalt	Malaysia	Oxisols	26–28°C	<250	500–2,000	4–24	4.0 to 5.3	(Anda et al. 2013)
Basalt	UK	clay-loam agricultural soil	N/A	50–150	10,000	2	6.6 to 6.7	(Kelland et al. 2020)
Dunite	Belgium	N/A	N/A	<43.5	22,000	12	6.6 to 8.5	(Amann et al. 2020)
Dunite	Belgium	N/A	N/A	<1020	22,000	12	6.6 to 8.0	
Olivine	Netherlands	Podzols	15−20° C	<200	163	8	4.89 to 4.99	(Ten Berge et al. 2012)
Olivine	Netherlands	Podzols	15−20° C	<200	815	8	4.89 to 5.10	

Olivine	Netherlands	Podzols	15−20° C	<200	4,070	8	4.89 to 5.34	
Olivine	Netherlands	Podzols	15−20° C	<200	20,400	8	4.89 to 5.96	
Olivine	Denmark	Podzols	20−25° C	20	5,000	3	3.56 to 4.69	(Dietzen et al. 2018)
Olivine	Denmark	Podzols	20−25° C	20	1,000	3	3.56 to 5.18	
Wollastonite	USA	Podzols	6°C	16	344	12	3.88 to 4.39	(Johnson et al. 2014)
Steel slag	China	Ultisol	25° C	<1000	5–26	9	5.82 to 6.68	(Ning et al. 2016)

2.3 Particle Size and Surface Area

Finer particles increase geometric surface area per unit mass, and mesocosm evidence and global modeling support higher near-term alkalinity generation under finer distributions and larger pH shifts for the same rock input (Amann et al., 2020; Baek et al., 2023; Beerling et al., 2020; Strefler et al., 2018; Taylor et al., 2016). Even so, three systematic deviations from the idealized picture must be accounted for. First, surface area does not scale simply with sieve size when micro-porosity, defects, and aggregation dominate; Brunauer–Emmett–Teller (BET) surface area, measured by gas adsorption, and geometric estimates can diverge. Second, the field-available surface area decays over time because selective erosion removes fines, while passivation and pit coalescence reduce active site density (Brantley et al., 2008; Pokrovsky and Schott, 2000; Calabrese et al., 2022; Tuğrul, 2004; White et al., 2017; Swoboda et al., 2022; Nguyen et al., 2016). Third, energy requirements for comminution grow rapidly as target size decreases, and life cycle emissions can overwhelm chemical gains if grinding is pushed too far (Rinder and von Hagke, 2021).

Model comparisons reveal that assumptions about grain size distributions, surface area scaling, shrinking-core behavior, and mineral specificity drive large differences in predicted removal at the same application rate (Baek et al., 2023; Strefler et al., 2018; Taylor et al., 2016; Beerling et al., 2020). Because these assumptions are only weakly constrained by field data, they should be treated as model structural uncertainty rather than as parameter noise. The operational inference is that net removal per unit energy has an interior optimum rather than a monotonic relationship with fineness. Identifying that optimum requires paired trials that report alkalinity fluxes and energy use, together with time-varying particle size distributions in situ.

2.4 Mineral Armoring and Coupled Reactions

Passivation layers that form through coupled dissolution–precipitation reactions reduce access of reactive fluids to fresh surfaces. In olivine-rich dunites and in wollastonite systems, thin Si-rich layers can precipitate when dissolved silica rises, although the degree of kinetic suppression differs across mineral structures, with wollastonite often less affected than forsterite (Daval et al., 2011; Amann et al., 2020). Reaction–diffusion treatments can replicate laboratory passivation, yet parameterization remains challenging in heterogeneous soils and is seldom included in large-scale ERW assessments, which likely biases removal upward when silica activities increase over time (Emmanuel, 2022; J. Weber et al., 2023; Baek et al., 2023; Kantzas et al., 2022).

Hydrologic conditions further modulate passivation. Under high rainfall, pore waters dilute, supersaturation declines, and passivation can weaken, which favors dissolution. Under drying, evaporation and limited flushing raise saturation ratios and encourage precipitation, which enhances armoring. Because reaction direction depends on saturation state, the balance between dissolution and precipitation varies with the residence time distribution and with transport pathways rather than being a fixed material constant (Brantley, 2008; Roelandt et al., 2010). For this reason, long-term field studies should co-monitor dissolved Si, cation release stoichiometry, and surface morphology to relate silica exposure histories to effective rate decay and to constrain model representations of passivation.

2.5 Environmental Conditions

Climatic controls are necessary but not sufficient. Natural weathering accelerates in warm, wet environments, and modeling studies often reproduce higher ERW potentials in the tropics and subtropics (Brady and Carroll, 1994; Levenson et al., 2017; Baek et al., 2023; Goll et al., 2021). Yet high rainfall can shorten fluid residence times and

truncate contact between pore waters and reactive surfaces, while riverine processing can partially negate upland alkalinity gains before they reach marine sinks (Klemme et al., 2022; S. Zhang et al., 2022). Therefore, climate-driven "fast kinetics" does not automatically translate into "more removal." The controlling variables are the joint distribution of residence times and saturation states along the flow path.

Soil properties and land management exert first-order influence on this joint distribution. Structure, connectivity, and cation exchange capacity determine infiltration, water holding, soil CO_2 profiles, and pore water chemistry, all of which reshape field-effective rates and the allocation of products among exchange sites, dissolved loads, and secondary phases (Deng et al., 2023; Johnson et al., 2014b; Maher and Chamberlain, 2014). Tillage, cover cropping, and harvest remove or redistribute cations and alter mixing depths, which in turn modify alkalinity budgets and transport (Beerling et al., 2020; Deng et al., 2023). Consequently, realistic assessments should be built around mass balance and flux closure rather than background descriptions of setting. A reasonable working expectation is that, at comparable climates, soil structural metrics that control residence time will explain more variance in net removal than climate variables alone. This expectation is testable with paired, cross-soil trials that combine tracer-informed hydrology with alkalinity export measurements.

3. Ecosystem Synergies of ERW

Biota modulate ERW not as a uniform accelerator but as a set of interacting controls on reaction thermodynamics, transport, and element budgets. Figure 1 places microbes, roots, and fauna alongside mineral surfaces and hydrologic pathways. The synergy becomes meaningful only when biological processes increase verifiable alkalinity export, maintain acceptable risk profiles, and avoid countervailing fluxes such as respiration-driven CO_2 release or nutrient losses. The sections below evaluate where evidence supports durable gains, where mechanistic claims rest on laboratory surrogates, and where field-scale flux closure remains unproven.

3.1 Soil Microbial

Microbial metabolism can increase mineral dissolution through acid production, biofilm-mediated attachment, and ligand-promoted complexation, and several studies document catalytic enhancement of rock weathering under controlled conditions (Blackmore et al., 2018; Pastore et al., 2022). However, much of the most detailed microbiological evidence comes from carbonate or mixed-lithology contexts, which limits direct transfer to silicate-dominated ERW unless alkalinity production and export are shown. Moreover, community composition and function vary with moisture, temperature, and N to P ratios, so climatic gradients confound simple extrapolation from single-site experiments (Z. Xu et al., 2018). Consequently, claims that microbial processes "accelerate ERW" require co-measurement of porewater alkalinity, cation stoichiometry, and residence times, together with attribution of CO_2 sources, since microbial respiration can increase pore-space pCO_2 while not necessarily contributing to net atmospheric drawdown. In practical terms, biofilm formation and organic acid secretion may increase near-surface dissolution yet also raise dissolved silica, which can promote surface passivation and lower effective rates over time; studies that report only early-time kinetics risk overstating sustained removal.

3.2 Plant Roots

Plant roots and mycorrhizal fungi amplify weathering by elevating pore-space CO_2, secreting organic acids, and intensifying exchange between solids and solutions, and these interactions are central to realistic ERW scenarios in managed and unmanaged systems (Deng et al., 2023; Dontsova et al., 2020; Swoboda et al., 2022). Organic ligands chelate metal centers and mobilize cations, thereby advancing dissolution beyond what proton attack achieves alone (Jones et al., 2009; Quirk et al., 2012). Yet the same biology that mobilizes cations also partitions them into biomass; if Ca and Mg accumulate in plant tissues and are removed at harvest, the alkalinity that would otherwise be exported can be curtailed, which affects net removal claims in croplands. In parallel, roots modify redox microenvironments by transporting oxygen and shaping electron acceptor distributions, which favors the formation of Fe and Si secondary phases that can stabilize soil organic carbon while simultaneously shielding fresh mineral surfaces from attack (van Veelen et al., 2020; Quirk et al., 2012). Therefore, studies that report biotic acceleration should also report matched cation budgets, biomass export, and evidence for downstream alkalinity transport, since enhanced dissolution at the root–mineral interface does not guarantee field-scale removal without mass-balance closure.

3.3 Soil Fauna

Soil fauna influence ERW primarily through physical restructuring of the medium rather than through direct catalysis. Burrowing, casting, and ingestion mix rock powders into reactive zones, fragment particles, and alter moisture and aeration, which can increase contact between porewaters and mineral surfaces (Osler and

Sommerkorn, 2007; Pinto-Figueroa et al., 2019; B. Tan et al., 2021; Xu Guo-Liang and Kuster, 2012; Morford et al., 2016). However, these same processes can also transport fines out of the targeted horizons and increase hydrologic connectivity, shortening residence times and potentially reducing the fraction of alkalinity that is retained long enough to be exported in solution. Evidence for fauna-driven acceleration is strongest at microcosm and plot scales and over short durations; few studies couple bioturbation metrics with time-resolved alkalinity fluxes and particle size distributions in situ. Until such data are available, the most defensible representation of fauna in ERW assessments is as modifiers of mixing depth, hydraulic residence time distributions, and reactive surface exposure, rather than as independent "boosters" of intrinsic dissolution kinetics.

3.4 Soil Nutrient Cycles and Terrestrial Ecosystem Carbon Cycles

Nutrient feedback offers a plausible path to durable ERW gains, but the direction and magnitude of the effects depend on soil chemistry and management. Increased pH from silicate dissolution can alter nitrification and denitrification pathways and has been proposed to lower N_2O and NO emissions in some settings, yet the sign of the response depends on moisture regimes, substrate availability, and microsite dynamics, so global-scale reductions should be treated as contingent rather than universal (Val Martin et al., 2023). Phosphorus dynamics provide a second axis of interaction. ERW can raise P availability through direct apatite dissolution and through enhanced mobilization of organic P mediated by microbial and root processes, which can shift plant uptake strategies toward greater mycorrhizal reliance and improve nutrient-use efficiency in P-limited systems (Goll et al., 2021; Bi et al., 2024). At the same time, not all silicate rocks provide substantial P, and pH shifts can precipitate P into less available forms in some soils, so lithology and background chemistry determine whether ERW delivers an agronomic dividend or a neutral outcome (Manning, 2018; Swoboda et al., 2022; te Pas et al., 2023).

Carbon cycle responses extend beyond dissolved inorganic carbon. Release of Ca, Mg, and Si can stabilize soil organic carbon through mineral–organic associations and aggregate formation, potentially increasing the fraction of SOC in more persistent pools, although documentable gains depend on the balance between enhanced inputs and any priming of decomposition following pH changes (Hartmann et al., 2013; Beerling et al., 2020; T. Xu et al., 2024). Because inorganic carbon formation and organic carbon stabilization have different permanence, monitoring frameworks should separately account for SIC and SOC, report mineral-associated organic matter fractions, and avoid conflating transient pH effects with long-term storage (Manning et al., 2024). Productivity responses add another layer. Field work indicates that ERW can raise net primary production in some systems, including bioenergy crops such as miscanthus, yet higher biomass does not automatically translate into greater net removal if harvest exports base cations or if hydrologic losses counterbalance added inputs (Kantola et al., 2023). Finally, model-based upper-bound estimates of carbon drawdown in P-limited regions, including values approaching 2 kg CO_2 m^{-2} yr^{-1} at application rates of 5 kg m^{-2}, are informative for potential but hinge on assumptions about limitation, stoichiometry, and flux closure that remain to be verified across soils and climates (Goll et al., 2021). For these reasons, robust claims about ecosystem synergies should be conditioned on co-reported alkalinity export, nutrient budgets, trace gas fluxes, and SOC fractionation, rather than on single proxies such as pH or short-term growth responses.

4. Evaluating ERW: Data, Lifecycle, Economics, and Policy Considerations

4.1 Data, Observations and Experiments

Laboratory kinetics for silicate dissolution are well established, yet pilot and field evidence for ERW spans more than four orders of magnitude, which directly propagates into uncertainty in carbon removal estimates (White & Brantley, 1995; White, 2008). **Table 3** makes clear that this spread is not random noise. The compilation spans croplands, grasslands, and forested catchments, and studies quantify different phases, including exchange sites, porewaters, plant uptake, secondary minerals, and adsorbed pools, and they do so under distinct hydrologic regimes, mineralogy, and soils. As a result, cross-study comparisons are only meaningful when measurements are anchored in mass balance with clear boundaries for what counts toward removal and when land-use context (agricultural vs. natural) is explicit. Where those boundaries differ, apparent performance differences are often methodological rather than mechanistic.

Even with these caveats, two patterns are consistent with first principles and with Table 1 across both agricultural fields and natural landscapes. First, mineral identity matters. Wollastonite applications repeatedly report higher removal and faster apparent weathering, which is congruent with its rapid dissolution under soil pH conditions. Second, pH exerts a strong control. Figure 2 shows that field-normalized rates for olivine vary by more than four orders of magnitude between pH about 3.5 and pH near 8, a spread that exceeds laboratory projections when fitted to standard rate laws. This divergence is expected if selective leaching and secondary precipitation increase with rising dissolved silica and if hydrologic contact times are insufficient to offset passivation. In that case, the relevant

comparison is not between laboratory lines and field points but between field points with similar saturation states and residence times.

Site heterogeneity further complicates inference in both managed (cropland) and unmanaged (e.g., forested watershed) systems. The Malaysian oil palm study illustrates how background practices and legacy amendments can mask a silicate signal. In that setting, widespread carbonate fertilization raised baseline weathering and alkalinity export, leaving little room for a detectable increment from silicate additions even under warm and wet conditions that should otherwise favor reaction (Larkin et al., 2022). Because such interactions are common in working landscapes, studies that lack controls on background alkalinity inputs risk conflating treatment effects with management history. As deployment scales up, these confounders will likely widen uncertainty bands unless experiments are designed to separate carbonate and silicate contributions across both croplands and natural areas.

Isotopic and elemental tracers offer a path to attribution. Strontium isotopes can distinguish silicate from carbonate sources; radiocarbon constrains dissolved inorganic carbon ages; rare earth elements and immobile element ratios can fingerprint mineral dissolution and secondary precipitation pathways (Peters et al., 2004; Larkin et al., 2022; Knapp et al., 2023; Kantola et al., 2023; Reershemius et al., 2023). This coupling is especially important in natural ecosystems where agronomic yield signals are absent, yet the value of these tracers depends on coupling them to hydrologic measurements and to cation and alkalinity budgets. Without that coupling, tracer-based assignments do not resolve whether observed dissolution results in exported alkalinity or is offset by within-soil reprecipitation or by harvest removal of base cations. Consequently, Figure 2 should be read not as a confirmation that laboratory kinetics mispredict field behavior, but as a prompt to include saturation indices, residence times, and tracer constraints in the same frame for both agricultural and natural sites so that field points can be conditioned on the variables that actually control divergence from laboratory expectations.

Table 3. Pilot- and field-scale observations of ERW.

Site/experiment description	Duration	Soil characterization	Feed stock characterization	Experiment condition	Measured phases	Total CO_2 removal (t CO_2/ha over duration)	Weathering extent[a]	Study
Planted mesocosm (PVC pots of 10 L, with top diameter of 25 cm)	32 wk (0.62 yr)	pH: 4.79 CEC[b] (cmol/kg): 19 Texture: sandy Source: Wageningen, Netherlands	Olivine SSA (N_2, BET)[c]: 4.8 m^2/g Particle size: P_{73} = 0.05 mm Application rate: 0.163 kg/m^2 (low); 20.4 kg/m^2 (high)	Olivine powder was fully mixed with soil; Natural rainfall; 400 ml water was supplied on top every 14 d; Duration: 32 weeks	Plant; Exchangeable	0.28 (low); 2.58 (high)	0.148 (low); 1.1×10^{-2} (high)	(ten Berge et al. 2012)
Flow-through reactor on soil cores (0.1×1 m)	5 mo (0.42 yr)	Calcareous soil Source: North Oxfordshire, United Kingdom	Olivine SSA (BET): 3.04 m^2/g Particle size: P_{65} = 0.212 mm Application rate: 12.7 kg/m^2	Olivine powder was applied to top 20 cm layer; Constant flow at 16.8 m/yr Temperature: 19 ˚C Duration: 5 months	Leachate; Exchangeable; Secondary carbonate	0.54	6.6×10^{-3}	(Renforth et al. 2015)
Incubation experiment (110 ml bottles)	3 mo (0.25 yr)	pH: 3.55 Texture: sandy Source: Nørholm	Olivine (Forsterite) Particle size: P_{50} = 0.02 mm	Olivine powder was fully mixed with soil;	Exchangeable	4.15 (high); 3.13 (low)	7.1×10^{-2} (high); 0.267 (low)	(Dietzen et al. 2018)

		Hede, Denmark	Application rate: 1 kg/m^2 (low), 5 kg/m^2 (high)	De-ionized water was added to maintain soil moisture; Temperature: 20-25 ˚C Duration: 3 months				
Planted mesocosm (rooftop pots of 9 L, with top diameter of 24 cm)	8 wk (0.15 yr)	pH: 4.94 Source: Ontario, Canada	Wollastonite SSA (N_2, BET): 20.28 m^2/g Particle size: P_{90} = 0.026 mm Application rate: 22 kg/m^2	Wollastonite powder was fully mixed with soil; Natural rainfall Duration: 8 weeks	Secondary carbonate	39.3	0.74	(Haque et al. 2019)
Planted mesocosm (length: 26 cm, diameter: 46 cm)	730 d (1 yr reported)	pH (KCl): 6.6, 7.8 CEC (cmol/kg): 8.6 Texture: loam-sand Source: Belgium	Olivine (dunite) from Almklovdalen peridotite complex SSA (N_2, BET): 9.5 m^2/g (fine), 1.1 m^2/g (coarse) Particle size: P_{80} = 0.04 mm (fine), 1.02 mm (coarse) Application rate: 22 kg/m^2	Olivine powder was applied to top 11 cm layer; Irrigated daily and weekly; Mean infiltration rate: 800 mm/yr	Leachate	0.049 (fine); 0.023 (coarse)	9.9×10^{-5} (fine); 4.6×10^{-5} (coarse)	(Amann et al. 2020)
Planted mesocosm (length: 50 cm, diameter: 16 cm)	120 d (0.33 yr)	pH (H_2O): 6.6 CEC (cmol/kg): 25.4 Texture: loam-sand Source: Leicestershire, United Kingdom	Columbia River Basalt SSA (N_2, BET): 7.35 m^2/g Particle size: P_{80} = 1.25 mm Application rate: 10 kg/m^2	Basalt powder was applied to top 25 cm layer; Irrigated every 5 d; Mean infiltration rate: 767 mm/yr Temperature: ~25 ˚C Duration: 120 d	Leachate; Plant; Exchangeable; Secondary carbonate	0.78	1.6×10^{-2}	(Kelland et al. 2020)
Soil core (0.1×1 m)	5 wk (0.10 yr)	Calcareous, loamy soil Source: North Oxfordsh	Columbia River Basalt SSA (N_2, BET): 16.3 m^2/g Particle size: P_{80} = 1.25 mm	Basalt powder was applied to top 13 cm layer; Natural rainfall	Leachate (the cores were flushed by 330 ml of	0.001	6×10^{-5}	(Buckingham et al. 2022)

		ire, United Kingdom	Application rate: 10 kg/m^2	(infiltration rate: 57 mm/yr) Duration: 5 weeks	rainwater before sampling)			
Planted mesocosm (length: 100 cm, diameter: 50 cm)	99 d (0.27 yr)	pH: 7.72 Texture: loamy Source: Belgium	DURUBAS basalt SSA (BET): 9.2 m^2/g Particle size: P_{75} = 0.25 mm Application rate: 5 kg/m^2	Basalt powder was applied to top 20 cm layer; watered regularly with tap water (infiltration rate: 143 mm/yr)	Leachate; Plant; Exchangeable; Secondary carbonate	The detected difference is smaller than soil heterogeneity		(Vienne et al. 2022)
Hubbard Brook Experimental Forest (11.8 ha)	1-11 yr	pH: 3.89 Texture: sandy loam Location: New Hampshire, USA	Wollastonite SSA (BET): 1.6 m^2/g Particle size: P_{50} = 0.016 mm Application rate: 0.345 kg/m^2	Wollastonite powder (pelletized to 1 to 3 mm) was applied by helicopter; MAP [d]: 140 cm Annual discharge: 87 cm MAT [e]: 6 ◦C Duration: 11 y	Leachate; Plant; Exchangeable	>1.24 (1 yr, river bank); > 1.76 t (11 yr, whole watershed)	> 0.635 (1 yr, river bank); > 0.924 (11 yr, whole watershed)	(Peters et al. 2004) (Green et al. 2013) (Johnson et al. 2014)
Farm (18 ha potato field)	5 mo (0.42 yr)	pH: 5.92 Texture: Sandy loam Location: Alliston, Canada	Wollastonite SSA (N_2, BET): 2.07 m^2/g Particle size: P_{50} = 0.014 mm Application rate: 0.124 kg/m^2	Applied using a lime spreader Irrigated Duration: 5 months	Secondary carbonate	0.32	0.68	(Haque et al. 2020)
Farm (2 ha soybean field)	20 wk (0.38 yr)	pH: 6.59 Texture: Sandy loam Location: Woodstock, Canada	Wollastonite SSA (N_2, BET): 3.48 m^2/g Particle size: P_{50} = 0.008 mm Application rate: 0.5 kg/m^2	Applied using a lime spreader Not irrigated Duration: 20 weeks	Secondary carbonate	0.40	0.21	
Oil palm plantation (0.6-1.8 ha)	3 yr	pH: 5-5.8 Location: Sabah, Malaysia	Pleistocene magmatic-arc volcanic deposits	MAP: ~2000 mm MAT: 28 ◦C	Leachate; Exchangeable;	No significant difference was detected.		(Larkin et al. 2022)

			SSA (N_2, BET): 2.73 m^2/g Particle size: P_{50} = 1.77 mm Application rate: 5 kg/m^2		Secondary carbonate			
Farm (3 × 20 m plots)	3 yr	pH (H_2O): 5.8; pH ($CaCl_2$): 5 CEC (cmol/kg): 7.16 Texture: clay-loam Source: Vojens, Denmark	Ilulialik rock flour SSA (N_2, BET): 19.6 m^2/g Particle size: P_{50} = 0.0026 mm Application rate: 5 kg/m^2	Ilulialik rock flour was applied to top 8 cm layer by harrowing; MAP: 897 mm MAT: 9.4 ◦C Duration: 3 y	Exchangeable	0.72	7.9×10^{-2}	(Dietzen and Rosing 2023)
Miscanthus plots (0.7-3.8 ha)	5 yr	Texture: silt loam Location: Illinois, USA	Blue Ridge basalt SSA (N_2, BET): 1.02 m^2/g Particle size: P_{80} = 0.276 mm Application rate: 5 kg/m^2	Applied using a lime spreader MAP: 1051 mm MAT: 11 ◦C Duration: 5 y Yearly application for 3 (miscanthus) and 4 years (maize/soybean)	The difference in total elemental composition in solid phases. Plant	43.0	0.29	(Kantola et al. 2023)
Maize/soybean plots (0.7-3.8 ha)	5 yr					18.5	0.09	

[a] *Weathering extent is quantified as the ratio of the mass of minerals that have undergone weathering to the total initial mass of minerals present, over the course of the experiment.* [b] *CEC: cation exchange capacity.* [c] *specific surface area (SSA) measured via N_2 adsorption and calculated using the Brunauer–Emmett–Teller (BET) method.* [d] *MAP: mean annual precipitation.* [e] *MAT: mean annual temperature.*

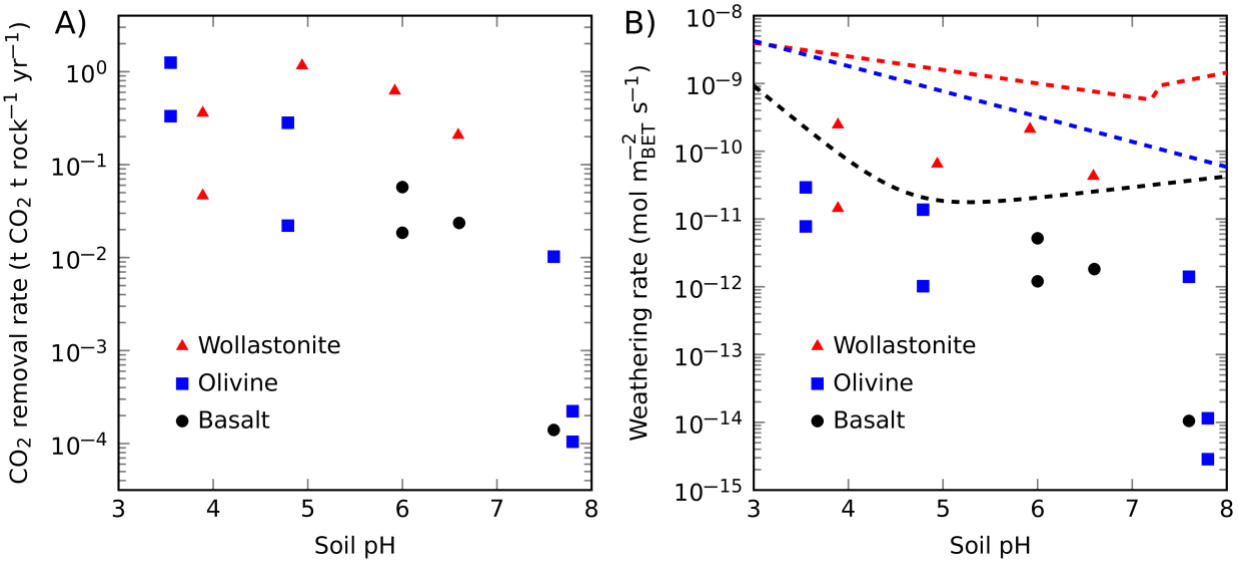


Figure 2. Relationships Between CO_2 Removal Rate, Weathering Rate, and Soil pH. The CO_2 removal rate (A, normalized to the initial amounts the applied minerals) and weathering rate (B, normalized to the initial surface area of the applied minerals) exhibit a distinct dependency on pH, as evidenced by pilot- and field-scale observations of Enhanced Rock Weathering (ERW) (**Table 3**) across agricultural plots and natural catchments. These rates (represented as points) underscore the critical role of pH in the efficiency of ERW processes. A comparative analysis with the rate law derived from controlled lab-scale measurements (illustrated as dashed lines fitted using the kinetics rate law reported in Bandstra & Brantley, 2008) highlights the potential variances in weathering kinetics under different environmental conditions.

4.2 Full Lifecycle of ERW

Figure 3 further reveals lifecycle CO_2 emissions, costs, and energy demands of ERW processes. Based on the data from Table S1 for each case study, we calculated the energy and emissions required to grind 1 tonne of rock. Grinding to finer particle sizes, such as 10 µm, results in emissions ranging from 0.18 to 163.55 kg of CO_2 and costs between $3.26 and $38.8, consuming approximately 0.002 to 1.2 GJ of energy (Figure 3). Those high variations are because the energy demands, emissions, and costs of ERW processes depend on factors like the hardness and mineral composition of the rock, regional electricity prices, and technological differences in grinding equipment across studies (Strefler et al. 2018; Kantzas et al. 2022; Li et al. 2024). Transport, particularly over long distances like 100 km, contributes emissions between 0.5 and 86.7 kg of CO_2, with costs ranging from $0.8 to $5.0. Spreading of crushed rock adds further costs and emissions, ranging from $10.38 to $171 and producing approximately 0.003 kg of CO_2. This phase often rivals transport in terms of emissions and costs, especially when specialized equipment or large-scale deployment is required. In contrast, mining and crushing activities have a lower impact, emitting around 4.29 kg of CO_2 and costing $5.46, while consuming 0.005 to 0.03 GJ of energy. These insights suggest that improving the efficiency of grinding, optimizing logistics to reduce transport distances, and minimizing the energy required for spreading could significantly lower the overall carbon footprint and economic costs of ERW.

Lifecycle analysis shifts the question from “can minerals dissolve” to “can they deliver net removal after energy and logistics are priced in.” for both agricultural and natural deployments. Table S1 and Figure 3 indicate that grinding dominates energy demand and emissions once particle sizes approach tens of micrometers, while mining and primary crushing are comparatively minor. Emissions and costs vary widely, which reflects differences in rock hardness and mineralogy, regional electricity prices and carbon intensity, and equipment performance across case studies (Kantzas et al., 2022; Li et al., 2024; Strefler et al., 2018). Transport is the second lever. Per kilometer emissions look small, but long hauls rapidly accumulate and can rival grinding when supply chains stretch beyond regional sources. Spreading can be either modest or material depending on method, with aerial application providing speed at the price of emissions and cost.

Two strategic implications follow. First, there is no universal “best” fineness across agricultural and natural settings. Because comminution energy rises steeply as target size falls and because field-effective surface area decays over time, net removal per unit energy has an interior optimum that depends on mineralogy, soil structure, and hydrology.

Identifying that optimum requires pairing lifecycle inventories with measured alkalinity export for the same materials, sites, and grain size distributions rather than relying on laboratory kinetics alone. Second, logistics should be designed backward from candidate sites and sinks. Co-siting crushing and pre-grinding with existing quarries or industrial facilities, leveraging existing farm and forest road networks, prioritizing short supply chains, and aligning deployment with seasons that maximize hydrologic contact (e.g., wet seasons in croplands and high-flow periods in forested headwaters) can reduce both costs and emissions. These choices are material to viability since the range in grinding and transport burdens shown in Figure 3 is comparable to, and sometimes larger than, the removal credited in field studies.

Finally, boundary conditions in lifecycle assessments must be explicit. Reported emissions should be tied to marginal rather than average grid intensities where relevant, capital amortization should be transparent, and coproduct credits, if any, should be justified rather than assumed, including explicit treatment of biomass harvest exports and land-management operations where applicable in natural systems. Without these conventions, cross-study comparisons collapse into accounting artifacts even when underlying engineering is sound. (Kantzas et al., 2022; Li et al., 2024; Strefler et al., 2018)

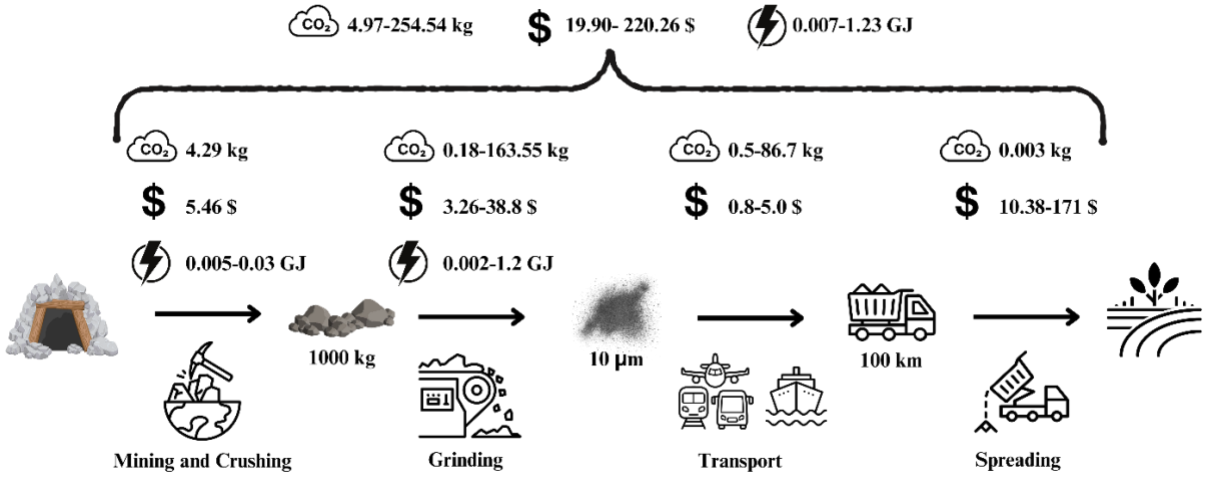


Figure 3. Lifecycle CO_2 emissions, costs, and energy demands of ERW processes applicable to both agricultural and natural deployments. This figure provides a visual representation based on the review of data from **Table S1**, which outlines the lifecycle CO_2 emissions, costs, and energy demands associated with key stages of the ERW process. These stages include mining and crushing, grinding, transport, and spreading. The example considers 1 tonne of rock being crushed to a particle size of 10 µm and transported over a distance of 100 km. The chart highlights the variation in energy input, emissions, and costs involved at each phase of the ERW process, supporting a comprehensive understanding of the environmental and economic impacts of ERW implementation.

4.3 Economic considerations

Economic viability rests on more than headline cost per ton. It depends on the joint distribution of costs for comminution, transport, and spreading, on the durability and measurability of removal, and on any agricultural co-benefits that persist once nutrient budgets and harvest exports are accounted for (Figure 4). In agricultural systems, co-benefits may include yield gains; in natural systems, benefits center on acid neutralization, base-cation replenishment, and water-quality improvements rather than crop yields. The mesocosm and field examples indicate that basalt applications can raise yields in some contexts, and multi-year trials in major cropping systems report double-digit increases in maize and soybean production, which can partially offset deployment costs (Kelland et al., 2020; Beerling et al., 2024). Yet yield gains are not fungible with removal credits unless the same experiments document alkalinity export and cation budgets. Otherwise, revenue stacking risks double counting benefits that draw on the same underlying processes.

Regional heterogeneity is the rule. Energy prices, labor, equipment access, and suitable lithologies vary across settings, which produces broad cost bands in practice and widens the uncertainty on scale-up trajectories (Doetterl et al., 2018; Smith et al., 2019; Strefler et al., 2018), and land-use constraints (e.g., access and permitting on public lands) can further shape feasibility in natural areas. Reuse of mining and industrial by-products can reduce feedstock costs and, in some cases, lower environmental liabilities, but quality control on trace metals and particle size distributions becomes essential for both agronomic acceptance and measurement, reporting, and verification (MRV) integrity (Moosdorf et al., 2014; Psarras et al., 2020; Goll et al., 2021; Stavins, 1999) with screening tailored to

sensitive natural habitats. A defensible economic case therefore links site-specific lifecycle costs to verifiable removal and to clearly separated agronomic benefits or non-agricultural co-benefits, and it expresses uncertainty through sensitivity to power prices, haul distances, and grain size targets rather than through point estimates.



Figure 4. Enhanced Rock Weathering Process and Impact Analysis (reflecting both agricultural and natural ecosystems). This infographic provides a comprehensive overview of the Enhanced Rock Weathering (ERW) process as an innovative strategy for climate change mitigation by removing atmospheric CO_2. The figure details each stage of the ERW process: mining and grinding rock into varying particle sizes for optimal dissolution; spreading these materials across agricultural fields and natural landscapes (e.g., grasslands, forested catchments); and the subsequent CO_2 absorption and transformation into stable soil carbonates and plant biomass. It further explores the agricultural and economic impacts of ERW. Agriculturally, ERW contributes to increased soil fertility through mineral dissolution, enhancing crop yields and long-term carbon sequestration capacity compared to depleted soils without ERW intervention. In natural systems, ERW can mitigate soil and stream acidification, replenish base cations, and support watershed health. Economically, the diagram breaks down initial investment versus long-term operational costs and discusses regional disparities in cost-effectiveness due to varying soil compositions and climates. The figure distinguishes co-benefits that are agriculture-specific (e.g., yield) from those relevant to natural systems (e.g., water-quality gains), avoiding double counting in MRV. Additionally, the infographic highlights the potential for substantial economic gains from increased crop yields where applicable and environmental benefits such as CO_2 reduction and improved soil health. Lastly, the figure flags land-use-specific risks and safeguards (e.g., trace metals, sediment loss, habitat disturbance) for both settings. Within this framework, the depicted activities are iterative, addressing feedbacks and interactions to promote the sustainability of the ERW process.

4.4 Policy Considerations

Policy determines whether ERW remains a collection of local experiments or evolves into a managed carbon removal option with standards that guard against over-crediting and environmental risk in both agricultural and natural landscapes. Jurisdictions with mature climate policy frameworks can integrate ERW into existing instruments if MRV protocols are strong enough to ensure that alkalinity export and permanence are credibly demonstrated and that heavy metal risks, soil health, and downstream water quality are kept within regulatory limits (Strefler et al., 2018; Smith et al., 2019; Lehmann and Possinger, 2020; Oppon et al., 2024) with distinct MRV modules for agricultural fields and for natural ecosystems (e.g., forests, rangelands, headwater catchments). In settings where resources and regulatory capacity are limited, policy emphasis may rationally shift toward soil enhancement and food security, particularly where croplands are extensive and P limitation is widespread, while in natural landscapes the emphasis may be on acid mitigation and watershed health, but this strategy still requires guardrails and support for measurement to avoid trading near-term agronomic gains for long-term environmental costs (Beerling et al., 2020; Schenuit et al., 2024; Mir et al., 2023; Renforth et al., 2009; Holzer et al., 2023).

Because climate benefits and hydrologic impacts cross borders and many natural headwaters are transboundary, international collaboration has a direct bearing on ERW standards. Harmonized definitions of removal, shared reference methods for alkalinity and tracer measurements, and interoperable data formats would improve comparability and reduce the risk of cherry-picking favorable accounting. Equity considerations also matter. Policies should ensure that the communities who host mining, processing, and spreading retain agency and gain tangible benefits, and that public health and education are integrated with deployment where appropriate (Cox et al., 2022; Lawford-Smith and Currie, 2017; Vandeginste et al., 2024) including communities adjacent to natural areas slated for ERW pilots. Figure 4 is thus most useful when read as a policy design checklist. It explicitly separates pH adjustment from alkalinity export and provides parallel risk indicators for agricultural and natural deployments (e.g., trace metals, sediment loss, habitat disturbance), and encourages regional planning that links feedstock supply to demand while keeping haul distances tractable. Effective frameworks will remain flexible, but they should be anchored in mass-balance-based MRV and in protections that sustain soil and water quality alongside carbon accounting and adapted to land-management regimes outside agriculture.

5. Modeling Perspectives in ERW

Table 4 organizes approaches by spatial and process scale. This is preferable to framing models as "process based" versus "biogeochemical" or "reaction transport," which obscures the fact that all tiers encode processes but resolve different constraints.

Table 4. Hierarchical Classification of ERW Modeling Approaches, Key Features, and Limitations.

Model types	Representative Models	Spatial/Temporal Scale	Key Strengths	Key Limitations	References
Reactive-transport	PFLOTRAN, TOUGHREACT, CrunchFlow	cm–m; hours–months	High-resolution geochemistry; detailed kinetics; explicit diffusion & precipitation modeling	High computational cost; limited biological processes; challenging to scale up or validate in field	(Pandey and Rajaram 2016). (Deng et al. 2023)
Soil & Ecosystem Biogeochemical	PROFILE, ForSAFE, 1D-ICZ, DMBM, Sheffield Model	m–km; seasons–decades	Integrates soil hydrology, plant interactions, and biological weathering;	Simplified mineralogy; uncertain parametrization; often site-specific calibration	(Warfvinge and Sverdrup 1992) (Kronnäs et al. 2019). (Giannakis et al. 2017; Kotronakis et al. 2017) (Taylor et al. 2016) (Cipolla et al. 2021, 2022)
Earth System Models	CESM–CLM5, cGENIE (w/ weathering modules)	Global; decades–centuries	Captures feedback with atmosphere–ocean; can simulate large-scale deployment and climate impact	Oversimplified terrestrial processes; instantaneous bicarbonate export assumption may overestimate sequestration	(Maffre et al. 2018) (Vakilifard et al. 2021) (Vandeginste et al. 2024) (Val Martin et al. 2023)

Integrated Assessment & Optimization Models	WITCH, GCAM, MILP logistics frameworks, c-RISE	Regional–global; multi-decadal	Incorporates costs, land competition, grid-carbon intensity, policy scenarios; aligns with CDR deployment needs	Relies on coarse assumptions from upstream models; rarely includes uncertainty bounds or mechanistic constraints	(Goddéris et al. 2006) (Zhang et al. 2023b) (Zhang et al. 2023a) (Morrow et al. 2023) (Kantzas et al. 2022)

Reactive transport models provide the most explicit representation of coupled chemistry and transport at pore to plot scales. They can track dissolution, precipitation, diffusion, and evolving saturation states, which makes them indispensable for evaluating hypotheses about passivation, silica feedbacks, and residence time effects on effective rates. Yet they often operate under boundary conditions that are easier to impose in a laboratory than to verify in a field setting, and they rarely include dynamic biological forcings such as root uptake and microbial production of ligands at relevant temporal resolutions (Pandey and Rajaram, 2016; Deng et al., 2023). Computational burden then limits ensemble sizes and the exploration of parameter non-uniqueness. For decision support, the clearest contribution of this tier is to produce calibrated, transferable relationships between effective rate, saturation index, and residence time that downstream models can use, rather than to deliver single-scenario forecasts that are difficult to validate. Taken together, these pore- to plot-scale constraints define effective rate–saturation–residence-time relationships that directly feed the next tier of soil and ecosystem models.

Soil and ecosystem models extend the domain to seasons and years and incorporate plant–water–soil interactions with explicit dynamics. They can represent mixing depths, infiltration and drainage, cation exchange, and biological enhancement of dissolution while tracking management interventions such as tillage and harvest. Their geochemical cores are necessarily simplified, and mineralogies are often reduced to a few classes, which introduces structural uncertainty that is easily hidden by site-specific calibration (Warfvinge and Sverdrup, 1992; Kronnäs et al., 2019; Giannakis et al., 2017; Kotronakis et al., 2017; Lyla L. Taylor et al., 2016; G. Cipolla et al., 2022; Giuseppe Cipolla et al., 2021). This tier is most informative when it reports mass-balanced alkalinity export, cation budgets that include plant uptake and removal at harvest, and distributions of residence times. Without that suite of outputs, agreements with pH or yield trajectories risk being diagnostic of agronomy rather than of carbon removal. Those mass-balanced exports and residence-time distributions then provide boundary conditions and priors for Earth system models.

Earth system models link land to river networks and the ocean, where permanence is ultimately established. They can explore teleconnections and policy-relevant scales, including the response of marine carbonate chemistry to sustained alkalinity inputs. Their strength is the representation of feedbacks between land, ocean, and atmosphere over decades to centuries. Their weakness is the tendency to assume rapid or instantaneous export of alkalinity from soils and rivers, which likely overstates sequestration when real flow paths include storage, re-precipitation, and biogeochemical attenuation (Maffre et al., 2018; Vakilifard et al., 2021; Vandeginste et al., 2024; Val Martin et al., 2023). The most constructive way to use this tier is to propagate calibrated fluxes and uncertainties from the terrestrial models, to include realistic routing and transformation in rivers, and to make the lag between application and ocean uptake an explicit, reportable quantity. The resulting long-horizon alkalinity and carbon trajectories inform integrated assessment and optimization models.

Integrated assessment and optimization models translate biogeophysical potential into deployment pathways. They evaluate source–sink logistics, power mix, land competition, and policy scenarios and are therefore essential for comparing ERW with other options within constrained portfolios (Goddéris et al., 2006; Y. Zhang et al., 2023; B. Zhang et al., 2023; Morrow et al., 2023; Kantzas et al., 2022). However, they sit at the end of the modeling chain and inherit every upstream simplification. When average removal rates, constant costs, and thin uncertainty bounds are supplied, large-scale projections reflect accounting choices more than physical limits. Their added value grows when they accept ensembles of removal and cost trajectories conditioned on grain size, residence time distributions, and life cycle burdens, and when they return the implied infrastructure and power demands back to the process tiers for feasibility checks. In short, each tier conditions the next, and feedbacks back to process-level feasibility need to be explicit.

As a bridge to the section 7, we summarize cross-tier implications and practical gaps below and then point forward to broader recommendations. Cross-tier recommendations: reactive transport models should commit to publishing effective-rate surfaces over saturation and residence time, with and without passivation. Soil and ecosystem models

should commit to alkalinity export closure with matched cation budgets and to sensitivity analyses that separate hydrologic controls from kinetic parameters. Earth system models should commit to explicit transit times and transformation in river networks and to the separation of alkalinity injection from oceanic storage. Integrated assessment models should commit to reporting the marginal power and logistics requirements implied by any scale-up and to representing life cycle emissions at the same grain-size and transport resolutions used by the process models.

Staged validation across tiers: validation against terrestrial CO_2 consumption by weathering in watersheds tests geochemical plausibility and hydrologic routing, as in the Sheffield model's comparison across 42 basins (Lyla L. Taylor et al., 2016). Validation of nitrogen gas emissions responses to pH in CESM–CLM5 tests a different set of processes that are relevant to co-benefits and trade-offs rather than to removal itself (Val Martin et al., 2023). Both exercises are valuable. Neither substitutes for validation of alkalinity export, travel time to the ocean, or the partitioning between dissolved and secondary solid phases along the flow path. A coherent validation program would align metrics across tiers so that what is verified at one scale becomes a conditioned prior at the next. A coherent validation program should align metrics across tiers so that what is verified at one scale becomes a conditioned prior at the next.

Data-exchange principles: two integration practices follow from this critique. First, pass data upward in forms that preserve physics. Provide distributions or response surfaces rather than single means. Track uncertainty as structural and parametric components so that downstream models do not collapse it into a single error bar. Second, pull data downward in forms that enable falsification. Let field programs inform residence time distributions, tracer fingerprints for carbonate versus silicate contributions, and the fraction of alkalinity that reaches river outlets. In that arrangement, the hierarchy in Table 4 becomes a set of coupled constraints rather than a set of disconnected tools.

Traceability requirement: claims about ERW potential should always be expressed jointly with the modeling choices that support them. Statements about deployment scale or system cost that are not traceable to choices about grain size targets, passivation treatment, hydrologic residence times, and life cycle burdens will not survive cross-tier scrutiny. Conversely, when those choices are explicit and validated at each tier, the hierarchy can deliver predictions that are both decision relevant and scientifically defensible.

6. Comparative Analysis and Risks of ERW

6.1 Comparison of ERW with other CDR Techniques

Comparative assessments of carbon removal options are most informative when system boundaries, baselines, and monitoring requirements are aligned. Under this framing, Figure 5 functions as a benchmark for sensitivity rather than a definitive ranking. The salient question is which assumptions need to hold for each option to deliver verifiable removal at stated scales and costs, and how robust those assumptions appear in real landscapes and energy systems.

ERW couples potential carbon removal with agronomic services. Reported mid-century ranges span approximately 0.2 to 4 Gt CO_2 per year at costs of about 50 to 513 US dollars per tonne, while late-century scenarios extend to values up to 95 Gt CO_2 per year with costs from roughly 24 to 600 US dollars per tonne (Pathak et al., 2022; Strefler et al., 2021). These figures are useful for exploring parameter space, yet they embed choices about grain size targets, passivation, residence times, and supply-chain emissions. Field evidence indicates that such choices materially affect outcomes, so ERW potentials are best treated as conditional on site chemistry and hydrology, as well as on MRV focused on alkalinity export and cation budgets (Beerling et al., 2020; Goll et al., 2021; Zhang et al., 2023).

Direct Air Capture offers location flexibility and comparatively straightforward accounting of captured CO_2, although energy intensity typically governs cost. Mid-century projections of about 0.5 to 5 Gt CO_2 per year at 100 to 300 US dollars per tonne are consistent with this trade-off; late-century estimates near 40 Gt CO_2 per year often carry wide cost bands that reflect constraints in low-carbon power and infrastructure (Ruthven, 2014; Zeman, 2014; Fuss et al., 2018; Smith et al., 2016). Afforestation and reforestation provide biodiversity and hydrologic co-benefits but store carbon over longer horizons and are sensitive to land and water competition. Reported potentials of roughly 0.5 to 3.6 Gt CO_2 per year by 2050 and up to about 10 Gt CO_2 per year by 2100 occur at relatively low average costs, provided sustained stewardship and appropriate species–site matching (Anderson et al., 2014; Farley et al., 2005; Lautenbach et al., 2017; Fuss et al., 2018; Pathak et al., 2022).

Bioenergy with Carbon Capture and Storage (BECCS) integrates energy production and storage but inherits land and nutrient constraints associated with biomass, alongside capture and transport requirements. Mid-century ranges of approximately 0.5 to 5 Gt CO_2 per year at 100 to 400 US dollars per tonne and late-century values up to about 20 Gt CO_2 per year are contingent on sustainable feedstock supply and competing land uses (Fuss et al., 2018; Strefler et al., 2021). Biofuels and soil carbon sequestration occupy different positions in the portfolio. Net outcomes for biofuels remain sensitive to crop choice, management, and coproduct accounting, which complicates generalization and may limit scalability where arable land is constrained (Holzman, 2008; Johansson et al., 2020; Kochsiek and

Knops, 2012; Tittmann et al., 2010; Searchinger et al., 2015). Soil carbon sequestration and biochar often appear lower cost and nearer term, yet both face monitoring and permanence challenges and show pronounced spatial variability. Literature ranges include several gigatonnes per year by 2050 and larger values by 2100 with wide cost bands (Lal, 2004; Smith, 2016; Woolf et al., 2010; Fuss et al., 2018; Brack and King, 2021; Pathak et al., 2022). Wood vaulting (WV) stores woody biomass in engineered, oxygen-limited and dry vaults to slow decomposition. Reported mid-century ranges are roughly 1–10 Gt CO_2 yr^{-1} at US$ 30–100 t^{-1} CO_2, while long-run scenarios average 10.1–12.4 Gt CO_2 yr^{-1} at US$ 10–50 t^{-1} CO_2 under optimized logistics (Zeng and Hausmann 2022; Zeng et al. 2024; Luo et al. 2025).

Taken together, ERW appears most compelling where soils are acidic, phosphorus limited, and hydrologically suited to move alkalinity to streams without substantial loss or reprecipitation. In such settings, soil services can offset part of the cost while MRV focuses on export rather than proxy variables. Where baseline carbonate inputs are common or residence times are short, observed gains are likely to be smaller unless management is adjusted. WV may complement these portfolios where woody residues are abundant and storage sites are proximate, offering comparatively low energy dependence but high siting and MRV dependence. Given National Academies targets near 10 Gt CO_2 per year by 2050 and about 20 Gt CO_2 per year by 2100, portfolio design benefits from combining options with less-correlated uncertainties and credible monitoring pathways at scale (National Academies of Sciences and Medicine, 2019; IPCC, 2022; Haszeldine, 2009; Wilcox et al., 2021; Montzka et al., 2011; Hoegh-Guldberg et al., 2019).

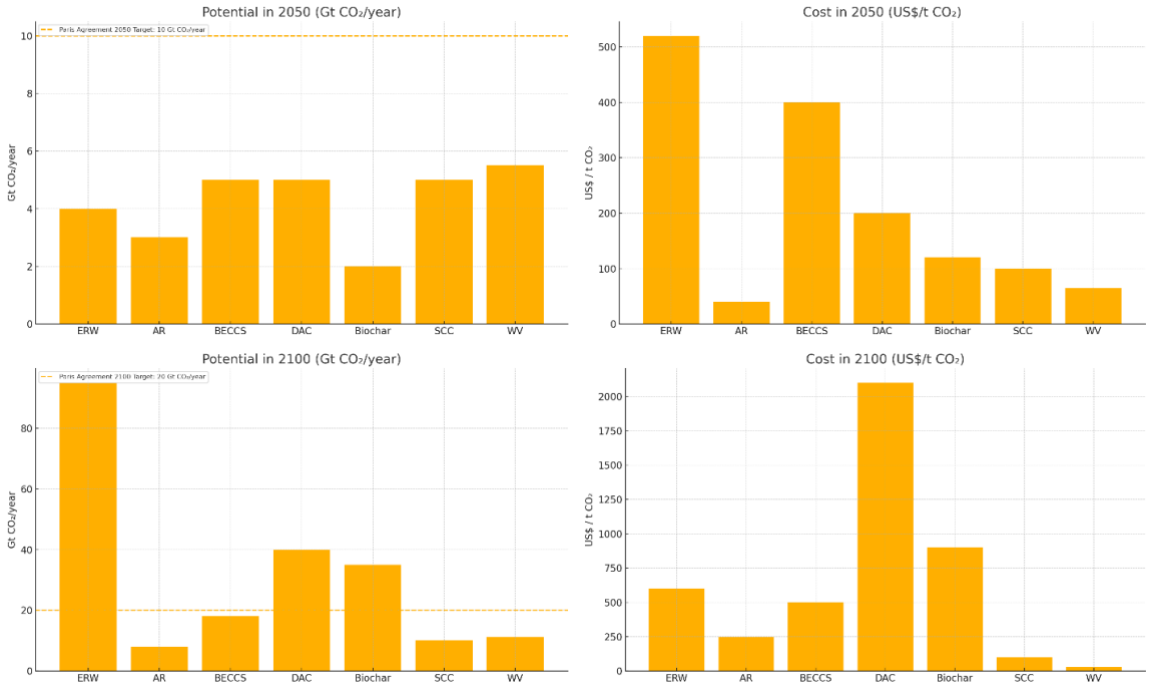


Figure 5. Ranges of potential and cost for carbon dioxide removal methods by 2050 and 2100. This figure is based on the data presented in Table S2. It compares the potential (in Gt CO_2/year) and cost (in US$/t CO_2) of several CDR technologies for 2050 and 2100. The CDR methods include: Enhanced Rock Weathering (ERW), Afforestation and Reforestation (AR), Bioenergy with Carbon Capture and Storage (BECCS), Direct Air Capture (DAC), Biochar, Soil Carbon Sequestration (SCC), and Wood Vaulting (WV). The orange boxes represent the projected range between the minimum and maximum potential and cost values for each method. The dashed orange lines represent the targets of 10 Gt CO_2/year by 2050 and 20 Gt CO_2/year by 2100, as estimated by the National Academy of Sciences (National Academies of Sciences and Medicine 2019), indicating the scale of CDR required annually to meet the Paris Agreement's goals. Projected cost ranges may rise in 2100 due to scale-related factors like land and resource constraints, even as technology matures.

6.2 Adverse Effects of ERW

Large-scale deployment invites attention to environmental health and social acceptance alongside carbon accounting. Trace metal mobilization is a recurring concern. Ultrabasic and some mafic feedstocks contain nickel and chromium that can be released during weathering, and field and mesocosm studies indicate that repeated or

heavy applications may raise soil concentrations toward regulatory thresholds within decade-scale horizons at higher application rates (Hartmann et al., 2013; Haque, Chiang, et al., 2020; Dupla et al., 2023). Metals often sorb to iron oxides, which can limit near-term leaching while establishing reactive pools that are susceptible to remobilization under reducing or highly acidic microsites. Observations in soils and waters implicate adsorption and subsequent transport pathways for copper, lead, chromium, manganese, and nickel (Dupla et al., 2023; Saikia et al., 2022; Okereafor et al., 2020). Feedstock screening, pre-application leaching tests, and post-application monitoring are therefore prudent program elements. Where risk is elevated by legacy contamination or sensitive receptors, phytoremediation with hyperaccumulator species has been proposed, ideally coupled with careful biomass handling to avoid secondary releases (Suhrhoff, 2022).

Soil chemistry responses point to additional trade-offs. ERW can raise pH and improve nutrient availability in acidic soils, yet in alkaline settings further pH elevation can reduce the availability of key nutrients and depress yields (Farhangi-Abriz et al., 2021). Short-term pulses in soil respiration and trace gas fluxes, including CO_2 and CH_4 in organic-rich soils, may erode near-term gains and complicate attribution if alkalinity export is not measured concurrently (Dupla et al., 2024; Beerling et al., 2018). Agronomic deployment is therefore better conditioned on baseline chemistry, buffering capacity, and crop removal of base cations, with adaptive management that responds to observed rather than assumed benefits.

Worker and community health risks arise mainly from dust during grinding, handling, and spreading. Fine particulates pose respiratory hazards, and the possibility of asbestos in some serpentine-bearing ultramafic rocks warrants screening and careful handling to meet occupational and public health standards (Vandeginste et al., 2024; Dostert et al., 2008). Practical mitigation includes feedstock characterization for fibrous phases, dust suppression, enclosed conveyance where feasible, appropriate protective equipment, and near-field air-quality monitoring during application.

Landscape and socioeconomic risks accompany efforts that aim for very large scales. Application across extensive areas can interact with other land uses, fragment habitats, and increase logistics burdens. Long-term costs for monitoring, risk management, and potential remediation remain uncertain and could challenge claims of low cost if carbon removal efficiencies vary across ecological and climatic gradients (Kantola et al., 2017; Beerling et al., 2020; Dupla et al., 2023; Vandeginste et al., 2024). Programs that align feedstock sourcing with existing quarries or industrial by-products, target soils where chemistry and hydrology favor alkalinity export, and embed monitoring from the outset tend to be more resilient than programs guided primarily by theoretical removal estimates.

Forward-looking perspectives and potential solutions. (i) QA/QC and dose design: adopt Ni/Cr “equivalent” caps tied to soil background and receptor sensitivity; screen for fibrous phases and pre-leach high-risk fines; report time-varying particle-size distributions. (ii) MRV that closes budgets: pair alkalinity-export accounting with cation budgets, saturation indices, and tracers (Sr isotopes, REEs) so treatment effects can be distinguished from legacy inputs. (iii) Hydrology-aware siting and timing: prioritize slopes/soils that maximize contact while limiting erosion; align spreading with wet-season infiltration windows; avoid wetlands and high-DOC hotspots that favor CH_4. (iv) Lifecycle minimization: co-site crushing with quarries/industrial plants, use low-carbon electricity, and target the interior optimum of “net removal per unit energy” rather than minimum particle size. (v) Occupational safeguards and social license: codify dust-control plans, buffer distances, and grievance mechanisms; share benefits with host communities via jobs, soil services, and monitoring access. These measures shift ERW from literature synthesis to deployable practice by linking site screening, MRV, and LCA to explicit guardrails and adaptive management.

7. Knowledge Gaps

7.1 Geological Inconsistencies and Unknowns

ERW performance varies with geology and geography, yet the spatial resolution and completeness of mineralogical maps remain uneven across regions, which limits siting and scaling analyses (Calabrese et al., 2022; Strefler et al., 2018). Mineral composition is recognized as a first-order control, although global identification of favorable lithologies is still incomplete. Soil properties then modulate outcomes through multiple pathways. Routine descriptors such as texture, pH, organic matter, and bulk cation exchange capacity capture part of the variance, while macroporosity, mineralogical assemblages, pore water chemistry, and cation exchange capacities disaggregated by ion and organic matter dependence often govern contact times and reaction trajectories yet are less frequently measured or mapped (Abdalqadir et al., 2024; Haque et al., 2019).

Hydrogeochemical context further conditions weathering rates. Position in the watershed, interactions between soils and shallow groundwater, bedrock weathering, and groundwater composition can shift saturation states and transport pathways, thereby altering apparent rates and the fate of alkalinity (Bailey et al., 2014; Velbel and Price, 2007; de Oliveira Garcia et al., 2020; S. Zhang et al., 2022). These interactions are not fully represented in present siting frameworks. Ecological feedbacks add another layer of uncertainty. Microbial communities, plant health, and water

quality can respond to ERW in ways that either reinforce or counteract intended geochemical effects, and these responses are likely to vary with setting (Edwards et al., 2017).

A priority is to link mineral resource mapping with soil functional traits and hydrogeologic domains at resolutions relevant to deployment. Doing so would allow models to move from lithology-only screening to coupled assessments that include contact times, saturation regimes, and ecological risk indicators.

7.2 Data Gaps and Experimental Shortcomings

Progress is constrained by fragmented observations and heterogeneous methods. Long-term field studies remain sparse, even though they are well suited to evaluate persistence of alkalinity export, soil health trajectories, and ecological responses (Edwards et al., 2017). Short experiments provide mechanistic insight but often lack the temporal coverage needed to separate transient pH adjustment from durable export. Multi-scale data are also limited; plot-scale findings are not consistently nested within catchment monitoring, which complicates attribution across spatial scales (Hall and André, 2001).

Methodological inconsistencies hinder comparison. Reported weathering rates draw on different phase inventories, and protocols for normalizing by surface area, distance from equilibrium, and residence time are not standardized, which reduces interpretability across studies (S. Weber and Martinez, 2017). Hydrologic measurements, including discharge, residence time distributions, and routing to streams, are frequently incomplete despite their central role in translating state changes into fluxes. Expanded use of sensor networks, tracer tests, and remote sensing would improve transport characterization, while coordinated designs can align geochemical sampling with hydrologic state variables (Baek et al., 2023; Giuseppe Cipolla et al., 2021; Strefler et al., 2018; S. Zhang et al., 2022). Baseline surveys of soils and geology are also uneven, which complicates detection of treatment effects where background inputs are substantial.

7.3 ERW Modeling Gaps

Several limitations recur across modeling tiers. Hydrologic and biogeochemical feedbacks are often simplified, even though moisture regimes, ion exchange, and biological exudates can alter reaction environments through time and across landscapes (Giuseppe Cipolla et al., 2021). Molecular-scale processes, including secondary precipitation and surface passivation, are commonly omitted or represented implicitly, which risks overestimating long-term dissolution (Dorn et al., 2013). Terrestrial models are rarely coupled to explicit river-network transport, so the fraction of alkalinity that reaches marine sinks remains uncertain (Maher and Chamberlain, 2014; Sulman et al., 2022).

Land-use dynamics present another gap. Changes in crop type, tillage, irrigation, and harvest practices modify mixing depths, dissolution environments, and logistics, while broader land-cover shifts alter hydrologic connectivity and residence times (Dwarakish and Ganasri, 2015; Garg et al., 2019; Turner, 2010). Supply chains and logistics are typically treated outside process models, yet source–sink distances, transport modes, and handling constraints influence both emissions and cost trajectories. Integrated frameworks that exchange information between process modules and logistics components would yield more realistic scenarios.

Linkages to empirical data can be tightened. Common benchmark datasets from laboratory and field trials would support systematic validation and intercomparison. Hybrid strategies that employ machine-learning emulators to represent expensive process modules offer a practical route to ensemble projections and uncertainty propagation, provided that emulators are trained within the envelopes defined by experiments. A modular, multi-scale, data-informed architecture would allow fine-scale kinetics to inform larger-scale predictions while maintaining transparency about uncertainties.

7.4 Environmental Safety and Socio-Economic Integration

Long-term environmental effects remain incompletely quantified. Soil chemistry, surface and groundwater quality, and broader ecosystem dynamics can respond in site-specific ways to ERW, and the influence of feedstock variability on these endpoints is not yet well characterized (Lewis et al., 2021; Edwards et al., 2017; de Oliveira Garcia et al., 2020; Eufrasio et al., 2022). Socio-economic implications for small-scale and subsistence farmers are also underexplored. Costs of access, labor demands, and infrastructure gaps may limit adoption, and regional differences in market institutions and resource availability suggest that one policy design will not fit all contexts (Val Martin et al., 2023; Beerling et al., 2020). Community perspectives and cultural factors likely affect acceptance and stewardship but are rarely integrated into study designs.

7.5 Policy Bottlenecks and Implementation Barriers

Lack of clear regulatory frameworks creates uncertainty for investors and implementers, while limited incentives reduce early adoption (Beerling et al., 2020). Public awareness and acceptance vary across settings, which affects siting and permitting (Pidgeon and Spence, 2017; Spence et al., 2021). Infrastructure for processing, transport, and application is uneven, and large programs would require coordinated build-out across sectors that do not typically interact (Moosdorf et al., 2014; R. R. Tan et al., 2022). Land tenure and land-use priorities can further constrain access, particularly where policies are restrictive or contested (Smith et al., 2019). Equity and environmental safety considerations are important for legitimacy and durability yet are not consistently embedded in policy design.

7.6 Measurement, Reporting, and Verification for ERW

Despite growing interest in ERW as a scalable CDR method, significant knowledge gaps persist in MRV, undermining accurate quantification and public trust (Clarkson et al. 2024; Beerling et al. 2025). One major gap involves methodological uncertainties in proxy-based approaches, such as using rare earth element (REE) concentration changes to estimate CDR rates, which may overestimate removal due to unaccounted soil heterogeneity and non-uniform dissolution; critiques highlight that ordinary least squares regression often fails to capture these variabilities, leading to inflated or unreliable results (Kantola et al. 2023; Wolf et al. 2024). Similarly, discrepancies between solid-phase (e.g., soil carbonates), aqueous-phase (e.g., bicarbonate in leachate), and gas-phase MRV methods remain unresolved, with solid methods potentially overlooking transient losses and aqueous methods underestimating due to incomplete sampling (Clarkson et al. 2024). Rock-specific factors, including composition, dissolution sequences, and interference from strong acids (e.g., from fertilizers), further complicate verification, as they can decouple cation release from actual CO_2 consumption without standardized corrections (Power et al. 2025). Downstream dynamics introduce additional gaps: dynamic river network models reveal spatial heterogeneity in carbon leakage, where bicarbonate export varies by watershed hydrology, yet integrated basin-scale MRV frameworks are lacking (Zhang et al. 2025). Time lags from cation sorption in soil columns and secondary mineral precipitation also delay verifiable bicarbonate export, creating mismatches between field weathering and long-term sequestration (Beerling et al. 2025). Bridging these gaps demands interdisciplinary protocols, multi-method integration, and field-scale trials to enhance MRV robustness.

8. Recommendations for Enhancing the Deployment, Safety, Equity, and Scalability of ERW

To improve the deployment and efficacy of ERW as a climate mitigation strategy, strategic interventions are required. These recommendations focus on addressing monitoring technologies, research frameworks, modeling approaches, environmental safety, policy support, and cross-sector collaboration while ensuring safety, equity, and scalability:

Develop Adaptive Monitoring Technologies: Invest in real-time, adaptive monitoring systems that accommodate geological and environmental variability while ensuring safety (Stubbs et al. 2022). These systems should integrate advanced geospatial tools, enhanced sensor networks, and remote sensing technologies to track hydrological, geological, and ecological changes at various scales (Calabrese et al. 2022). A global mapping of mineral (Maus et al. 2020) and soil properties (Chen et al. 2022) is essential for identifying safe and effective ERW sites, ensuring that potential environmental risks are minimized. This data will also support the equitable distribution of ERW benefits, as areas with varying geological contexts can be better understood and managed for long-term sustainability.

Establish Large-Scale, Adaptive Research Frameworks: Implement large-scale, adaptive research frameworks that support both long-term data collection and immediate action. These frameworks should combine geospatial analysis with comprehensive soil, hydrological, and geological surveys, targeting key regions with scalable ERW potential (Beerling et al. 2020). The introduction of pilot projects in diverse ecological and agricultural settings will provide critical, context-specific data on ERW's impacts, ensuring safe practices and equitable access to its benefits (Levy et al. 2024). Additionally, focusing on small-scale and subsistence farming communities will help ensure that ERW strategies are not only scalable but also accessible and fair (Haque et al. 2021).

Integrate Holistic and Inclusive Modeling Approaches: ERW strategies require the integration of comprehensive modeling frameworks that account for hydrological cycles, biogeochemical processes, and nanoscale reactions while also projecting long-term carbon sequestration dynamics. Models should incorporate land-use changes (Zeng et al. 2022), logistical challenges (Renforth 2012), and socio-economic factors (Strefler et al. 2018) to ensure that ERW practices are both scalable and equitable. Special attention must be given to accurately representing environmental safety concerns and ensuring that ERW implementation does not disproportionately affect vulnerable communities. These models should also address the logistical complexities of large-scale rock transport, which is crucial for ERW's scalability and effectiveness (Tan et al. 2022).

Assess and Mitigate Environmental Risks: Conduct thorough evaluations of the environmental risks associated with ERW, particularly its effects on soil health, water quality, and ecosystems, across diverse geological and ecological contexts (Lawford-Smith and Currie 2017). The variation in risk between different rock types and geological conditions must be explored to ensure safety (Vandeginste et al. 2024). Developing localized, robust risk mitigation strategies will be essential to proactively address any negative environmental consequences. Ensuring that these strategies are equitable and do not disproportionately impact vulnerable regions or communities is critical for the responsible and sustainable scaling of ERW (Oppon et al. 2024).

Implement Comprehensive Policy and Regulatory Support: Governments and international organizations should develop clear regulatory frameworks and provide financial incentives to support safe and equitable ERW deployment (Webb 2020). This includes implementing MRV systems that ensure ERW projects adhere to safety standards and contribute effectively to carbon sequestration goals (Smith et al. 2020). Policymaking should prioritize public awareness, infrastructure development, and international collaboration to foster ERW's global scalability (Spence et al. 2021). Special attention should be given to the inclusion of marginalized and under-resourced communities in these frameworks, ensuring equitable access to the benefits of ERW.

Strengthen Academia-Industry Collaboration: Establish stronger partnerships between academic institutions and industry to accelerate innovation and ensure that research aligns with the practical needs of ERW deployment, particularly regarding safety and scalability (Power et al. 2023). Collaborative platforms should facilitate the exchange of data, insights, and best practices, reducing the time lag between academic discoveries and field implementation (Spence et al. 2021). Securing policy support and financial incentives for research and pilot projects, alongside addressing logistical challenges, will be key to scaling ERW while ensuring that the benefits are distributed equitably and safely across different sectors and regions (Cox et al. 2022).

By addressing these interconnected aspects with a focus on safety, equity, and scalability, ERW can become a more effective, scalable, and responsible solution for global climate mitigation efforts. Expanding research, fostering cross-sector collaborations, and developing supportive, inclusive policies will be pivotal to ensuring ERW's successful and sustainable integration into agricultural and environmental practices.

9. Concluding Remarks

ERW occupies a promising, yet contingent, position in the carbon removal portfolio. Its capacity to couple alkalinity generation with agronomic services creates opportunities in settings where soils and hydrology support export to durable sinks. At the same time, geological heterogeneity, data sparsity, methodological inconsistencies, and modeling simplifications introduce uncertainty that is best addressed through coordinated measurement, standardized protocols, and integrated modeling. Environmental safety and distributional considerations will shape public acceptance and policy support, particularly where feedstock chemistry, dust, and land use interact with local priorities. A practical path forward links siting to mineral and soil maps, aligns experiments with flux closure and hydrologic measurements, develops modular and data-informed models, and embeds equity and risk management into policy and program design. Under these conditions, ERW can contribute alongside other approaches to carbon removal and soil restoration. Realizing that contribution depends on evidentiary standards that travel from plot to basin to ocean and on institutions that support sustained monitoring and adaptive management.

Authors' Contributions

J.M. conceived the idea for the article. Y.H. performed the literature search and data analysis. J.M. and Y.H. drafted the manuscript. All authors contributed to the critical revision of the work and approved the final manuscript.

Statements and Declarations

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The authors declare that no new data were used for created in producing this manuscript. Figures 1 and 4 were created using Adobe Photoshop and Adobe Illustrator. Figure 2 was created using PGF (a graphing package in latex). Dataset used to create Figure 2 can be directly derived from Table 3 of this paper, and Bandstra and Brantley (2008). Figures 3 was created using Canva and is based on data from Table S1.

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